

QueueFlow: Computer Vision Based Automated Queue Ticket Assignment and Real-Time Predictive System for Dynamic Campus Service Delivery

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APPROVAL SHEET

This capstone research of **ARCHIE D. BALBIN** and **JOSHUA ROVIC P. SAN-OTA** entitled “**QUEUEFLOW: COMPUTER VISION BASED AUTOMATED QUEUE TICKET ASSIGNMENT AND REAL-TIME PREDICTIVE SYSTEM FOR DYNAMIC CAMPUS SERVICE DELIVERY**” was prepared and submitted in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science. It has been examined and is recommended for acceptance and approval for oral examination.

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Abstract

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Queue management in Philippine higher education institutions remains predominantly manual, resulting in inefficient counter assignments, inconsistent service monitoring, and prolonged student wait times during peak periods. This paper presents QueueFlow, a Smart Queue Management System that uses You Only Look Once version 8 nano (YOLOv8n) with object tracking, OpenCV camera processing, automated queue-number assignment, ticket generation, and real-time wait-time prediction. The implemented prototype separates the detector from the web backend: a camera-connected detector process performs YOLO inference, filters duplicate and implausible detections, estimates crowd metrics, and pushes tracked-person metadata to a FastAPI backend. The backend confirms new queue entrants through a configurable queue zone, candidate confirmation window, motion checks, track-ID remapping, and hybrid appearance-spatial re-identification before issuing a queue number. The intended deployment issues physical queue tickets through a thermal printer; for prototype demonstration, the system temporarily generates PDF tickets containing the same queue number, QR code, short code, and JSON Web Token (JWT). Ticket records are stored in MySQL, and students can check live status through a mobile browser or mobile tracker. The staff dashboard displays live annotated snapshots, queue state, no-show alerts, counter controls, analytics, and forecasts. Prediction outputs include current estimated wait time and 5, 15, and 30 min forecasts using an M/M/c wait-time baseline, short-term trend projection, Holt's double exponential smoothing, and a mean-reversion long forecast. Developed for Naga College Foundation, Inc. (NCF) in Naga City, Philippines, QueueFlow is evaluated in terms of queue assignment accuracy, re-identification reliability, ticket and token validation,

prediction accuracy using MAE and MAPE, system latency, and user satisfaction.

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Chapter 1

Introduction

1.1 Background of the Study

Campus service delivery depends not only on the availability of staff and counters, but also on how quickly an office can recognize demand as it forms. During enrollment, payment, clearance, and other academic transactions, students often gather faster than service personnel can manually count, organize, and update the queue. This creates a practical problem for both sides of the service process: students have limited visibility of their waiting status, while staff have limited real-time information for deciding when to serve, skip, recall, or prepare additional service capacity.

At Naga College Foundation Inc. (NCF), queue congestion is not simply a matter of long lines. It affects the continuity of student transactions because campus offices such as enrollment, cashiering, and library services are part of a connected workflow. A delay in one service point can influence the next step a student needs to complete. The Commission on Higher Education recognizes accessibility and efficient service delivery as important indicators of institutional quality [1]; therefore, improving queue visibility is also part of improving the overall student service experience.

Conventional queue systems help arrange people in order, but they usually begin only after a student manually asks for a number or after staff encode the transaction. These systems can record issued tickets, yet they do not automatically observe the actual service area, confirm new arrivals from a camera view, prevent duplicate queue entries, or estimate how the line may change in the next few minutes. Earlier queue management tools also tend to focus on static numbering rather than combining automated detection, staff monitoring, student status access, and prediction in one workflow [2, 3].

QueueFlow is proposed in this study as a computer vision based queue management prototype for campus service offices. Instead of relying only on manual queue entry, the system uses a camera-connected YOLOv8n and OpenCV pipeline to detect persons inside a defined queue zone. YOLO is suitable for this task because it performs real-time object detection in a single network pass [4], and YOLOv8n provides a lightweight model that can be used in practical prototype settings [5]. QueueFlow extends detection with duplicate filtering, candidate confirmation, and hybrid re-entry matching so that queue records remain more consistent when people move or briefly leave the camera frame.

The queue ticket component of QueueFlow is designed for thermal printer deployment, allowing a physical queue ticket to be issued after a queue number is assigned. Because the prototype is demonstrated without the final printer hardware, the current implementation generates a PDF ticket as a temporary substitute. This PDF carries the same operational information expected from the thermal ticket, including the queue number, short code, and QR link for student status checking.

Beyond ticket generation, QueueFlow connects the detector, queue records, staff dashboard, and prediction module through a FastAPI-based backend and MySQL database. The prediction module supports current, 5-minute, 15-minute, and 30-minute wait-time estimates using queuing logic and short-term forecasting. Through this design, the study aims to provide NCF with a prototype that improves queue visibility, supports staff decision-making, and demonstrates how computer vision and predictive analytics can be applied to campus service management in a Philippine educational institution.

The Review of Related Literature therefore examines the technologies and concepts that support each part of the problem identified above. Computer vision and YOLO literature address automated queue observation, queue management literature explains ticketing and identity maintenance, and forecasting literature supports the wait-time estimates needed for staff decision-making.

1.2 Review of Related Literature

This section reviews related studies according to the major needs identified in the Background of the Study. First, it discusses computer vision for recognizing people in service areas. Second, it reviews YOLO as the detection approach used by the prototype. Third, it examines queue management systems, especially the limitations of manual and static ticketing. Finally, it presents predictive analytics as the basis for transforming live queue records into current and near-future wait-time estimates.

Computer Vision for Real-Time Person Detection. Computer vision has emerged as a transformative technology for automated crowd monitoring in institutional settings, enabling non-intrusive, real-time crowd density estimation and behavioral pattern recognition. Modern deep learning approaches process video streams to extract meaningful crowd metrics including density maps, movement trajectories, and occupancy levels across designated zones without compromising individual privacy [6]. Kumar et al. [7] demonstrated that deep learning-based crowd monitoring systems achieve superior accuracy even when handling occlusions and varying lighting conditions common in institutional environments. Park et al. [8] explored lightweight model architectures achieving processing speeds of 25–30 fps on embedded GPU platforms. Privacy-preserving techniques—including pose estimation without facial recognition, immediate anonymization at the camera level, and encrypted transmission of aggregate statistics—are recommended for deployment in public-facing institutional service areas [9]. These principles directly inform QueueFlow’s design, which processes bounding box data and colour histograms without storing or transmitting raw video.

You Only Look Once (YOLO). YOLO represents a breakthrough in real-time object detection by processing entire images in a single forward pass through the neural

network. Unlike conventional detection methods that rely on region proposals or sliding windows, YOLO’s unified architecture simultaneously predicts bounding boxes and class probabilities, enabling systems to detect and count people with exceptional speed and accuracy [4]. The objectness score for each detection is:

$$\text{Confidence}(\text{box}) = \text{Pr}(\text{Object}) \times \text{IoU}(\text{pred}, \text{truth}) \quad (1.1)$$

where $\text{Pr}(\text{Object})$ is the probability that an object exists in the cell and IoU represents the intersection over union between predicted and ground truth boxes [5].

Queue Management Systems. Queue management systems in institutional settings have evolved from physical ticket dispensers and manual counting to digital platforms with varying degrees of automation. Camera-based queue monitoring approaches offer a fundamentally different model by passively detecting and tracking all persons who enter a service area, eliminating the manual entry requirement entirely [10]. Re-identification of returning persons is a technically challenging aspect of camera-based systems requiring robust person tracking beyond simple frame-to-frame bounding box matching [11].

Predictive Analytics for Service Demand Forecasting. Predictive analytics transforms queue management from reactive to proactive by forecasting demand patterns and identifying potential bottlenecks before they materialize. The M/M/c queuing model, as established in the foundational queueing theory literature, enables calculation of key performance metrics including average wait times, queue lengths, and system utilization [12]. System utilization is expressed as:

$$\rho = \frac{\lambda}{c\mu} \quad (1.2)$$

where λ is the arrival rate, μ the service rate, and c the number of active servers. The M/M/c model processes real-time arrival rates from computer vision systems to calculate immediate wait times using Little’s Law:

$$W = \frac{L}{\lambda} \quad (1.3)$$

For medium-term forecasting, Holt’s double exponential smoothing tracks both queue level and trend with separate smoothing parameters [13]. For the 30 min forecast exposed by the prototype, growth-ratio and mean-reversion logic compare early versus recent queue history to detect systematic demand changes.

1.3 Synthesis of the State-of-the-Art

The literature reviewed in this chapter supports the central concern identified in the Background of the Study: campus service offices need better real-time visibility of queue conditions, not merely a list of issued numbers. Studies on computer vision show that camera-based person detection can reduce dependence on manual observation by capturing activity in a service area as it happens. YOLO-based approaches are especially

relevant because they provide the speed needed for live monitoring, making them suitable for a prototype that must observe students entering and moving through a defined queue zone.

However, the reviewed studies also show that detection is only the first layer of a complete queue management system. A person detector can identify bounding boxes, but it does not automatically solve the operational problems described in the Background: duplicate entries, temporary disappearance from the camera view, skipped queue positions, or student access to queue status. This makes tracking, deduplication, candidate confirmation, and re-entry matching necessary additions when computer vision is applied to actual service-office queues.

Research on queue management and predictive analytics further reinforces the need to move beyond static ticket numbering. Traditional queue tools can arrange students in order, but they provide limited support for staff decisions when demand changes quickly. Queuing theory and short-term forecasting address this limitation by translating arrival rates, active counters, and queue length into wait-time estimates. Multi-horizon prediction is therefore useful because staff need both immediate queue status and near-future indicators for counter planning.

Taken together, the reviewed works suggest that an effective campus queue solution should combine four functions in one workflow: camera-based queue entry, reliable queue identity management, ticket and status access for students, and predictive information for staff. These findings connect directly to QueueFlow’s design, which integrates YOLOv8n detection, hybrid re-entry matching, thermal-ticket-oriented queue number generation with a PDF demonstration fallback, a staff dashboard, and current, 5-minute, 15-minute, and 30-minute wait-time estimates.

Despite these advances, existing studies remain limited when viewed against the specific needs of NCF’s campus service context. Many focus on crowd counting, general queue applications, or prediction models as separate components. Fewer works demonstrate how these components can operate together in a Philippine educational institution where queue visibility, staff action, student status checking, and practical deployment constraints must be handled in a single prototype.

1.4 Gap Bridged by the Study

This study addresses a specific practical gap: there is currently no integrated system that combines passive camera-based queue entry using YOLO, hybrid multi-tier person re-identification, and multi-algorithm multi-horizon wait time prediction, specifically designed for campus service delivery in a Philippine educational institution context.

QueueFlow responds to this gap by developing these components in a unified architecture: a YOLO-based person detection and tracking pipeline with a custom NMS pre-filter and post-frame deduplication layer, a hybrid re-entry matching algorithm that combines spatial proximity and appearance similarity, thermal-ticket-oriented queue number generation with a PDF demonstration fallback, and prediction models exposed as current, 5, 15, and 30 min wait-time estimates.

1.5 Theoretical Framework

The theoretical framework of this study is anchored on the queue visibility problem described in the Background of the Study. Campus service offices need a way to observe student arrival, maintain queue order, estimate waiting time, and present useful information to both staff and students. QueueFlow addresses this problem through the combined application of computer vision, queuing theory, and predictive analytics.

Computer Vision Theory provides the basis for replacing manual line observation with automated person detection. In QueueFlow, this theory is applied through YOLOv8n and OpenCV, where camera frames are processed to detect persons inside a defined queue zone. YOLO’s single-pass detection approach supports real-time monitoring because it can identify objects quickly within each frame [4]. For the study, computer vision supplies the first layer of queue visibility: it allows the system to recognize possible queue entrants without requiring every student to manually register at a counter.

Queuing Theory, as formalized in classical queueing literature [12], explains how arrivals, service capacity, and waiting lines behave in a service system. The M/M/c model represents a multi-counter service environment in which the arrival rate, service rate, and number of active counters influence queue length, utilization, and waiting time. Together with Little’s Law, this theory provides the mathematical basis for estimating waiting time, queue length, and counter utilization from real-time arrival and service rates. In QueueFlow, queuing theory is connected to the system through queue records, active counter settings, and service-time assumptions. Once the computer vision layer identifies queue activity, queuing theory provides the formulas for estimating how long students may wait and how staff decisions about counters may affect service flow.

Predictive Analytics and Forecasting Theory extends the system from observing the present queue to anticipating near-future demand. The Background of the Study emphasizes that queue conditions change throughout the day, especially during enrollment and payment periods. For this reason, QueueFlow uses short-term forecasting with queuing logic, trend projection, smoothing, and mean-reversion principles to produce current, 5-minute, 15-minute, and 30-minute estimates. These forecasts support staff decision-making by giving the dashboard more than a static queue count; they provide an early indication of whether congestion is likely to increase or decrease [14, 13].

Together, these theories form the foundation of QueueFlow’s design. Computer vision converts camera input into observable queue activity; queuing theory converts queue activity and counter capacity into service metrics; and predictive analytics converts those metrics into near-future wait-time estimates. The resulting information is delivered through the staff dashboard, student status access, and queue ticket workflow, where the intended deployment uses a thermal printer and the current prototype uses PDF tickets as a demonstration fallback.

1.6 Conceptual Framework

The conceptual framework adopts the Input-Process-Output-Outcome (IPOO) model to illustrate how raw camera data is transformed through the QueueFlow pipeline into

actionable queue management intelligence. The *Input* consists of a live camera feed, queue-zone settings, counter configuration, service-time assumptions, and staff actions. The *Process* involves YOLOv8n detection, NMS pre-filtering, zone-based filtering, hybrid re-entry matching, candidate accumulation over a minimum confirmation window, queue state management including no-show detection and position recalculation, ticket generation, and multi-horizon prediction. The *Output* delivers a real-time queue list with assigned numbers, ticket credentials, individual wait times, and prediction cards to both the staff web dashboard and the student mobile portal. The *Outcome* is improved queue visibility, reduced manual ticket handling, and better staff decision support.

1.7 Statement of the Problem

This study seeks to address the following problems:

1. What are the current limitations of the manual queue management process at NCF service offices during peak demand periods?
2. How can a YOLO-based passive detection pipeline with hybrid re-identification reliably assign and maintain queue numbers without student interaction?
3. To what degree of accuracy can the implemented prediction models forecast current, 5 min, 15 min, and 30 min wait times?
4. How do service staff and administrators evaluate QueueFlow in terms of usability, detection reliability, ticket generation, and prediction usefulness?

1.8 Assumptions

Several assumptions underlie the proposed study. First, that the camera hardware provides sufficient video quality and coverage of the queue zone to enable reliable YOLO person detection under the lighting conditions and pedestrian densities typical of NCF service areas. Second, that YOLO’s tracking mechanism, supplemented by the system’s spatial re-identification layer, produces sufficiently stable person identities across the queue duration. Third, that the three prediction models generate forecasts that are actionable and sufficiently accurate for staff scheduling and counter management decisions. Fourth, that the thermal printer can be configured in the deployment area; where printer hardware is unavailable, PDF ticket generation can demonstrate the same queue-number, QR-code, and token workflow. Fifth, that service staff and administrators possess basic digital literacy to operate the web dashboard and interpret the system’s outputs.

1.9 Scope and Delimitation

This study focuses on the design, implementation, and evaluation of QueueFlow at NCF in Naga City, Philippines, covering the development of a YOLOv8n-based detection pipeline, a hybrid re-entry matching algorithm, a multi-horizon prediction engine,

a FastAPI-based REST API, a MySQL-backed data layer, and a ticket generation module designed for thermal-printer deployment. The current demonstration build generates PDF tickets with QR codes as a temporary substitute for printed thermal tickets. The system is deployed as a single-queue-zone, single-service instance. It does not currently support simultaneous multi-zone or multi-department management, priority queue differentiation based on student status or urgency factors, direct integration with the Campus Management System or Student Information System, or IoT sensor integration beyond the camera feed and queue ticket printer. The evaluation period is restricted to one semester of post-implementation operation.

1.10 Significance of the Study

The most immediate beneficiaries of this research are the service staff and students of NCF. Service staff gain a real-time queue dashboard that eliminates manual counting and provides data-driven insight into current queue length, individual wait times, and demand forecasts across three time horizons—enabling proactive counter management rather than reactive response to visible crowding. Students benefit from a passive system that requires no action on their part: upon entering the monitored area, a queue number is assigned and immediately accessible through any mobile browser using their access token.

For institutional administration, QueueFlow provides queue performance records that support evidence-based decisions about staffing, service hours, and physical layout. For the academic community, the study advances the integration of computer vision, person re-identification, ticket generation, and multi-algorithm predictive analytics in service queue management.

1.11 Definition of Terms

Access Token. A ticket credential assigned to each confirmed queue number. In the implemented prototype, this consists of a short code and a JWT-backed validation record that allows the student to check queue status without a login account.

Candidate Accumulation. A buffer mechanism requiring a new detection to appear consistently for a minimum confirmation window before a queue number is assigned, preventing transient detections from generating spurious tickets. The prototype uses a configurable default of 14 confirmation frames.

Holt’s Double Exponential Smoothing. A forecasting method based on exponentially weighted moving averages [13] used for the 15 min prediction, applying separate smoothing parameters for queue level ($\alpha = 0.4$) and trend ($\beta = 0.2$) with damping factor 0.85 to avoid over-projection.

M/M/c Model. A foundational multi-server queueing model from classical queueing theory [12] used for short-term (5–15 min) wait time prediction, calculating $W = L/\lambda$ from Little’s Law using real-time arrival rates from the YOLO detection pipeline.

Non-Maximum Suppression (NMS). Both a standard IoU-based NMS pass and a custom X-column/Y-adjacency filter applied to raw YOLO outputs before they reach

the queue tracker, suppressing duplicate head and body boxes YOLO commonly emits for a single person.

Queue Zone. A configurable rectangular region defined by four pixel coordinates on the camera frame; only persons whose bounding box centre falls within this region are tracked and assigned queue numbers.

Thermal Queue Ticket. The intended physical output of the system containing the queue number, QR code, and short code. In the prototype demonstration, the same ticket information is generated as a PDF while thermal printer hardware is not yet attached.

Re-Entry Matching. A hybrid algorithm that first matches a new detection to a missing person by spatial proximity, then by appearance histogram similarity, and finally by recency as a last resort.

You Only Look Once (YOLO). The YOLOv8n model with `persist=True` tracking, configurable input size, and confidence threshold, used to detect and track persons entering the queue zone in camera frames.

Chapter 2

Review of Related Literature

2.1 Computer Vision for Real-Time Person Detection

The application of computer vision to person detection and crowd monitoring in institutional settings has matured significantly over the past decade. Modern deep learning architectures process video streams to extract crowd metrics including density maps, movement trajectories, and occupancy levels without compromising individual privacy [6]. Park et al. [8] demonstrated that lightweight model architectures optimized for edge devices achieve 25–30 fps on embedded GPU platforms, making real-time crowd analytics feasible within the budget constraints of educational institutions. Beyond simple counting, modern systems track individual trajectories through service areas, identifying dwell times at service counters and queue formation patterns [15].

Privacy-preserving techniques such as pose estimation without facial recognition, camera-level anonymization, and transmission of aggregate statistics address ethical concerns inherent in camera-based monitoring in educational environments [9].

2.2 YOLO-Based Tracking and Deduplication

YOLO’s single-pass detection architecture delivers the speed necessary for real-time queue monitoring, processing entire frames in one forward pass to simultaneously predict bounding boxes, confidence scores, and class labels [4]. The YOLOv8 series advances this through improved multi-scale feature pyramid networks and anchor-free detection heads, achieving state-of-the-art accuracy-speed tradeoffs [5].

The duplicate detection problem—where YOLO emits multiple bounding boxes for a single person—is particularly pronounced in queue environments where people face the camera at close range. Custom geometric tests using X-column overlap and Y-adjacency thresholds have been proposed as supplements to standard NMS [11], forming the basis of the pre-filter implemented in QueueFlow’s detection pipeline.

2.3 Queue Management and Re-Identification

Camera-based queue monitoring enables passive, continuous tracking of service user arrival order without requiring any student interaction [10]. The core challenge beyond detection is person re-identification: when a tracked individual temporarily disappears from view and reappears with a new track ID, the system must recognize them as the same person and restore their original queue position rather than assigning a new number [11].

Re-identification approaches fall into three categories: appearance-based methods using colour histograms or deep feature embeddings, spatial-temporal methods using last known position and time elapsed, and hybrid methods combining both signals [16].

2.4 Predictive Analytics for Service Demand Forecasting

Multi-horizon forecasting provides both immediate and strategic insights by generating predictions at different time scales, each matched to a specific operational decision type [14]. The M/M/c queuing model provides analytical wait time calculations from real-time arrival and service rate measurements, making it suitable for current wait-time estimates and near-term tactical predictions [12].

For the 15 min range, Holt’s double exponential smoothing [13] extends simple exponential smoothing by separately tracking queue level and trend. Damping the trend component prevents over-projection during transient spikes, a practical necessity in campus service environments where queue lengths can surge and dissipate quickly. For the 30 min forecast implemented in QueueFlow, mean-reversion logic is used to limit long forecast uncertainty.

Chapter 3

Methodology

3.1 Research Design

This study utilized a Mixed-Methods and Developmental research design to systematically develop and evaluate QueueFlow. Quantitative data focused on technical performance metrics such as queue number assignment accuracy, re-entry matching accuracy, prediction model MAE and MAPE across three forecasting horizons, and real-time processing latency. Qualitative data was gathered through semi-structured interviews exploring the experiences and perceptions of service staff and administrators.

3.2 Population and Locale

The study was conducted at Naga College Foundation, Inc. (NCF) located in Naga City, Camarines Sur, Philippines. NCF was selected as the research site because it serves large volumes of student service transactions across enrollment, cashiering, and library operations, with pronounced peak periods that create the queue management challenges QueueFlow is designed to address.

3.3 Research Instrument

The study utilized a researcher-made structured questionnaire employing a five-point Likert scale to measure respondents' level of agreement with statements related to system usability, detection reliability, prediction usefulness, dashboard clarity, and overall satisfaction as shown in Table 3.1.

3.4 Ethical Considerations

This study strictly adheres to ethical standards to protect participants' rights, ensure data privacy, and maintain research integrity. Participation was entirely voluntary and informed consent was obtained from all participants. All personal information was

Table 3.1: Likert Scale Interpretation

Scale	Category	WM Range	Interpretation
5	Strongly Agree	4.21–5.00	Excellent
4	Agree	3.41–4.20	Very Satisfactory
3	Neutral	2.61–3.40	Satisfactory
2	Disagree	1.81–2.60	Poor
1	Strongly Disagree	1.00–1.80	Very Poor

anonymized and stored in password-protected systems. The study complied with the Data Privacy Act of 2012 (Republic Act No. 10173).

3.5 Statistical Treatment of Data

Data gathered was processed using both descriptive and technical performance metrics. Descriptive analysis used the Weighted Mean ($WM = \Sigma(f \times w)/N$), Overall Weighted Mean, and Percentage to summarize survey responses. System performance was evaluated using:

$$MAE = \frac{\sum |y_{\text{predicted}} - y_{\text{actual}}|}{n} \quad (3.1)$$

$$MAPE = \frac{1}{n} \sum \left| \frac{y_{\text{actual}} - y_{\text{predicted}}}{y_{\text{actual}}} \right| \times 100\% \quad (3.2)$$

calculated separately for each of the three prediction model horizons.

3.6 System Design and Architecture

QueueFlow is implemented as a FastAPI-based web application with a separate camera detector process, a MySQL backend, staff dashboard pages, and public queue-status endpoints for students as shown in Table 3.2. The detector process captures frames through OpenCV `VideoCapture`, runs YOLOv8n tracking with configurable image size and confidence threshold, filters duplicate and implausible bounding boxes, extracts crowd and prediction metrics, and sends tracked-person metadata to the backend. The backend then performs queue-zone filtering, candidate confirmation, track-ID remapping, re-entry matching, queue-state management, and ticket generation. The intended ticket output is a thermal queue ticket; the demonstration build uses generated PDF tickets with QR codes as a temporary substitute.

3.7 Software Development Process

The system was developed iteratively using the Agile development methodology, organized into sprint-based cycles covering: YOLO pipeline setup and NMS pre-filter development; re-entry matching algorithm design and debugging; no-show detection imple-

Table 3.2: System Tools and Technologies

Category	Technology	Rationale
Detection	YOLOv8n	Lightweight person detection with persistent tracking
Backend	FastAPI	REST API, dashboard routes, detector upload endpoints, and MJPEG streaming
Video	OpenCV (cv2)	Camera capture, frame preprocessing, annotation, and JPEG encoding
Tracking	Custom Python	Queue-zone filtering, track remapping, re-entry matching, and no-show handling
Prediction	Custom Python	M/M/c baseline, short trend projection, Holt smoothing, and mean reversion
Ticketing	Thermal printer; demo PDF	Physical queue ticket output, with PDF/QR substitute for demonstration
Database	MySQL	Queue records, short codes, JWTs, events, and transaction history
Auth	Session/JWT	Staff session login, camera-token validation, and student ticket tokens

mentation; prediction model development and accuracy tuning; FastAPI REST API and staff dashboard development; thermal-ticket workflow design with temporary PDF ticket generation for demonstration; and student queue-status integration testing.

3.8 Validation and Testing Methods

System validation was conducted through four primary approaches. *Alpha Testing* was conducted internally by the developers to identify and resolve functional issues. *Beta Testing* released the system to a limited group of service staff at NCF to observe real-world interaction and collect feedback. *Usability Evaluation* employed the structured Likert-scale questionnaire administered to service staff and administrators after a minimum of two weeks of regular system use. *Performance Testing* evaluated processing latency, prediction accuracy using MAE and MAPE against ground-truth wait times, re-entry matching accuracy verified against direct observation logs, and ticket flow validation from queue assignment to QR/short-code status lookup.

Chapter 4

Synthesis Matrix of Related Literature

Table 4.1 presents a synthesis matrix summarizing the reviewed literature, the methodology of each study, key findings, and the gaps identified that motivated the present work.

Table 4.1: Synthesis Matrix of Related Literature

Author/s	Year	Topic/Focus	Findings	Gap/Limitation
Zhang et al. [6]	2023	CV crowd density monitoring	Non-intrusive crowd metrics without privacy compromise	No queue-specific deduplication or re-identification
Kumar et al. [7]	2022	Person detection in occluded environments	High counting accuracy in crowded scenarios	No queue number assignment or re-entry matching
Park et al. [8]	2023	Lightweight vision for edge deployment	25–30 fps on embedded platforms	No application to queue management
Redmon & Farhadi [4]	2018	YOLO real-time object detection	Simultaneous bounding box and class prediction	No queue-specific deduplication
Jocher et al. [5]	2022	YOLOv8 multi-scale detection	State-of-the-art accuracy-speed tradeoff	Duplicate boxes in close-range scenarios
Alahmari & Alharbi [10]	2021	Camera-based queue monitoring	Non-intrusive continuous queue tracking	No hybrid re-identification; no multi-horizon prediction
Chen & Guo [16]	2023	Mobile queue status display	Reduced physical congestion through mobile monitoring	No YOLO-based passive entry
Brown et al. [14]	2022	Multi-horizon queue forecasting	Three-tiered prediction supports tactical decisions	Not integrated with live YOLO feed
Shortle et al. [12]	2018	Foundational queueing theory and M/M/c	M/M/c and Little’s Law enable wait-time calculation	No integration with CV inputs
Holt [13]	2004	Exponential smoothing for trend forecasting	Weighted moving averages capture level and trend	No queue-specific application

Bibliography

- [1] Commission on Higher Education (CHED). *Policies and Guidelines on Outcomes-Based Education and Institutional Sustainability Assessment*. Tech. rep. Quezon City, Philippines: Commission on Higher Education, 2022.
- [2] M. S. Shah and V. A. Madan. “Queue Management in Educational Institutions: A Review of Current Practices and Future Trends”. In: *International Journal of Modern Education and Computer Science* 10.7 (2018), pp. 32–41.
- [3] R. Chen and L. Gao. “A Real-Time Data Driven Approach for Adaptive Server Scheduling in Service Centers”. In: *European Journal of Operational Research* 308.2 (2023), pp. 586–600.
- [4] J. Redmon and A. Farhadi. *YOLOv3: An Incremental Improvement*. arXiv preprint arXiv:1804.02767. 2018.
- [5] G. Jocher, A. Chaurasia, and J. Qiu. *Ultralytics YOLOv8*. GitHub. [Online]. Available: <https://github.com/ultralytics/ultralytics>. 2022.
- [6] H. Zhang, R. Liu, and X. Zhang. “A Real-Time System for Crowd Density Monitoring Using YOLO and Depth Estimation”. In: *Journal of Visual Communication and Image Representation* 90 (2023), Art. 103704.
- [7] A. Kumar, P. Singh, and R. Sharma. “Deep Learning-Based Crowd Monitoring with Attention Mechanisms for Occluded Environments”. In: *IEEE Transactions on Image Processing* 31 (2022), pp. 2140–2152.
- [8] J. Park, S. Kim, and D. Lee. “Lightweight Architectures for Real-Time Crowd Analytics on Edge GPU Platforms”. In: *Sensors* 23.10 (2023), Art. 4890.
- [9] L. Bergman and K. Nakano. “Privacy-by-Design Computer Vision for Institutional Crowd Monitoring”. In: *IEEE Security & Privacy* 21.3 (2023), pp. 34–43.
- [10] A. A. Alahmari and S. R. Alharbi. “Smart Queue Management System Using Computer Vision and Deep Learning”. In: *Electronics* 10.23 (2021), Art. 2951.
- [11] S. Rodriguez and K. Hernandez. “Design of a Vision-Based Service Time Estimation and Queue Alerting System”. In: *Sensors* 24.1 (2024), Art. 135.
- [12] John F. Shortle et al. *Fundamentals of Queueing Theory*. 5th ed. Hoboken, NJ: John Wiley & Sons, 2018. ISBN: 978-1-118-94352-6. DOI: 10.1002/9781119453765.
- [13] Charles C. Holt. “Forecasting Seasonals and Trends by Exponentially Weighted Moving Averages”. In: *International Journal of Forecasting* 20.1 (2004), pp. 5–10. DOI: 10.1016/j.ijforecast.2003.09.015.

- [14] R. Brown, T. Adams, and S. Lee. “Multi-Horizon Forecasting for Service Demand in Campus Environments”. In: *Expert Systems with Applications* 215 (2022), Art. 119402.
- [15] T. Nguyen, L. Tran, and H. Pham. “Individual Trajectory Tracking Through Service Areas Using Computer Vision”. In: *Pervasive and Mobile Computing* 95 (2023), Art. 101657.
- [16] Y. Chen and L. Guo. “Real-Time Monitoring and Prediction of Queue Length Using Computer Vision and LSTM Networks”. In: *Multimedia Tools and Applications* 82.31 (2023), pp. 40751–40772.

Curriculum Vitae

Appendices